

Orbital and Stellar Parameters of Symbiotic Stars

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Abstract. This paper reviews current knowledge of symbiotic binaries, with special emphasis on their multifrequency observational characteristics and the basic parameters of the symbiotic system components. We start with a brief presentation of variable phenomena found in symbiotic stars. This is followed by a summary of recent progress in the determination of their orbital and stellar parameters. We also discuss the basic properties of symbiotic giants compared to single evolved giants, as well as the nature of the hot component and its outburst evolution.

1. Introduction

Symbiotic stars were isolated as a separate class of spectroscopically peculiar stars by Annie J. Cannon during her work on the HD Catalogue at the beginning of the XX century. A typical symbiotic spectrum should show (Figure 1) absorption features—such as TiO bands and neutral metals—and an associated red continuum typical for red giants; a blue continuum with the H I Balmer lines and a jump in emission; and strong emission lines from relatively highly ionized species—He I, He II, [O III], etc.—usually found in planetary nebulae. The simultaneous presence in a single object of low temperature absorption features and emission lines that require high excitation conditions apparently points to its binary nature. In fact, binary models for symbiotic stars, in which a rather normal M giant was accompanied by a hot component resembling the central star of a planetary nebula, were proposed and discussed soon after their discovery (e.g., Berman 1932). Berman also proposed that the observed eruptive behavior of some objects could be caused by some kind of the hot-component instability, perhaps similar to that responsible for the nova phenomenon. It is really impressive how close these first models were to the generally accepted present-day model.

According to this model, a typical symbiotic binary consists of an M-type giant transferring material to a much hotter white dwarf companion via a stellar wind. The wind is ionized by the hot component giving rise to a symbiotic nebula. In some systems the red giant is replaced by a yellow G–K-type giant, and the white dwarf by a main sequence or neutron star. Most symbiotic stars (~80%) contain a normal giant and their near-infrared (NIR) colors show the presence of a stellar photosphere, $T_{\text{eff}} \sim 3000\text{--}4000$ K; these are classified as S-type (stellar) systems. The remaining 20% of symbiotic systems contain Mira-type variables, and their NIR colors are consistent with the combination of a

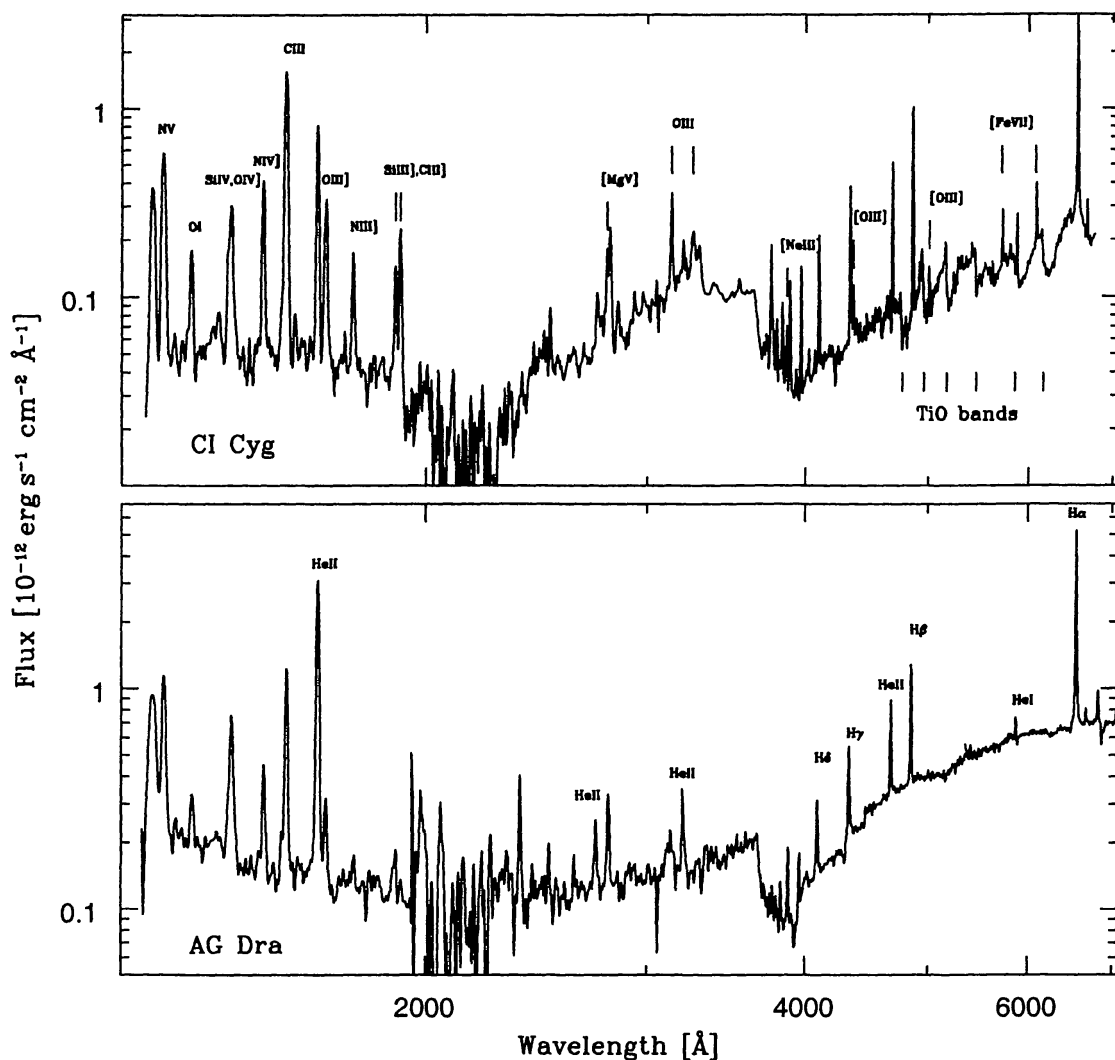


Figure 1. Quiescent optical/ultraviolet spectra for symbiotic stars. Strong TiO bands in the red spectrum of CI Cyg, indicate an M4–5 III primary; they are absent in the yellow, Galactic halo system AG Dra with a K3 II–III primary. The intensity of the UV continuum and He II emission lines require a similar temperature, $\gtrsim 10^5$ K, and luminosity, $\sim 1000 L_{\odot}$, of the hot component in both systems, whereas the CNO emission lines in AG Dra are systematically weaker than those in CI Cyg, which reflects the low metallicity of AG Dra. Another difference is the presence of strong high excitation forbidden lines (e.g., [Fe VII], [Mg v], and [O III]) in CI Cyg that are not visible in AG Dra.

reddened Mira and warm ($T \sim 1000$ K) dust shell; these are classified as D-type (dusty) systems.

Since the presence of an evolved red giant is indispensable for making a symbiotic binary, the system must have enough space for such a large star. Symbiotic stars are thus interacting binaries with the largest orbital separations, and their study is essential for understanding the interactions of detached and semi-detached binary stars. Mass accretion onto the hot component also plays a very important role in driving the basic properties and evolution of symbiotic stars, and involves energetic phenomena relevant to many other astrophysical

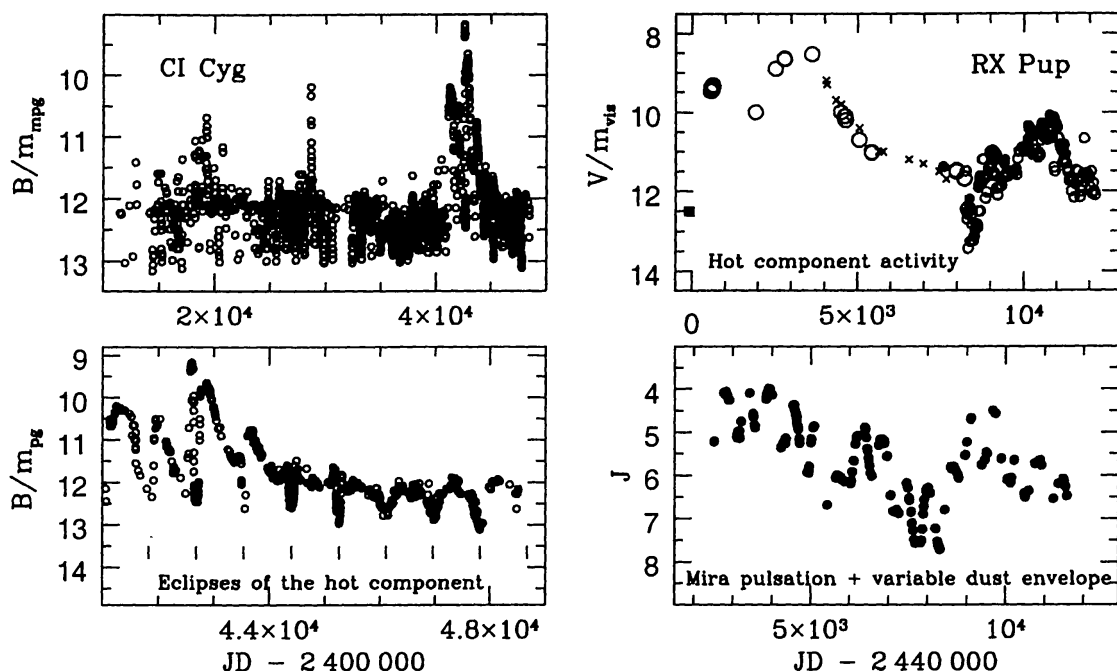


Figure 2. *Left:* The hot component outbursts (*top*) and deep eclipses (*bottom*) are the most prominent features of the 1894–1991 B/m_{pg} light-curve of the classical S-type symbiotic system CI Cyg. *Right:* The optical light-curve (*top*) of the D-type system RX Pup is dominated by the hot component activity, whereas the NIR light (*bottom*) is dominated by the Mira pulsation and the variable obscuration of the Mira by its circumstellar dust.

objects. The presence of both an evolved giant with heavy mass loss and a hot companion copious in ionizing photons also results in a rich and luminous circumstellar environment surrounding the interacting stars.

Such a complex multicomponent structure makes symbiotic stars a very attractive laboratory for studying various aspects of stellar evolution in binary systems. It is worth mentioning here the very interesting relatives of symbiotic stars: GX 1+4—consisting of an M6 III giant and a neutron star companion, and the peculiar black hole binary GRS 1915+105 recently found to comprise a K-giant companion (Greiner et al. 2001), which is among the most exotic and variable kinds of X-ray binaries. It is very important to set firm links between symbiotic stars and related objects in order to understand the role of these binaries in the formation of stellar jets, planetary nebulae, novae, supersoft X-ray sources, and SNe Ia. Many of these issues concern the late stages of stellar evolution and are presently poorly known but have important implications for our understanding of stellar pulsations and the chemical evolution of galaxies, as well as of the extragalactic distance scale.

The goals of this presentation are to review recent progress in the determination of the orbital and stellar parameters of symbiotic stars and to provide input for a more detailed discussion of such important topics as the basic properties of the symbiotic cool giant components compared to single evolved giants, the nature of the hot component and the physics of its outbursts, the origin of symbiotic circumstellar nebulae and their links with planetary nebulae, and the

evolutionary paths leading to symbiotic stars and to their progeny. We start with a brief description of the variable phenomena found in symbiotic stars in Section 2. Recent progress in the determination of their orbital and stellar parameters is summarized in Section 3. I discuss the basic properties of symbiotic giants compared to single evolved giants (note that symbiotic Miras are discussed in detail by Whitelock, this volume, p. 41) in Section 4, as well as the nature of the hot component and its outburst evolution in Section 5.

2. Light-Curves and Variability

The composition of symbiotic stars, specifically the presence of a red giant and its accreting companion, places them among the most variable of stars. They can fluctuate in many different ways. In addition to periodic changes due to the binary motion, both the cool giant and the hot component can also show intrinsic variability (Figure 2). For the red giant, these can be radial pulsations (all D-type and some S-type systems) and semiregular (S-type) variations of the giant, with timescales of the order of months and years, as well as long-term light changes due to variable obscuration by circumstellar dust (mostly D-type), solar-type cycles, etc. Mass accretion onto the hot component also results in different variable phenomena with timescales ranging from seconds and minutes (flickering and QPO) to years and decades (nova-like eruptions). Examples of light-curves for well-studied, though not yet completely understood, systems are RX Pup, CI Cyg, and CH Cyg; these are representative of the different kinds of variabilities observed in these objects and are also presented and discussed by Mikołajewska (2001).

The wealth of variable phenomena found in symbiotic stars is a real challenge for patient observers in any spectral range and makes these systems excellent targets for long-term monitoring programs, especially since they do not need large telescopes. This also explains how and why symbiotic systems are so effectively elusive in their physical nature. In fact, the binary nature of all symbiotic stars was definitely confirmed only in the 1980s by both direct measurement of the binary motion and the direct detection of the hot-component continuum in the ultraviolet spectral range, whereas various problems concerning their intrinsic variability and its relation to their binary nature remain unresolved.

3. Orbital Parameters

Although regular photometric changes, with periods of 200–1000 d and amplitudes increasing towards short wavelengths, had already been detected in many symbiotic stars in the 1930s, they were not generally associated with binary motion. The first symbiotic stars announced to be eclipsing binaries were AR Pav (Mayall 1937) and CI Cyg (Hoffleit 1968).

Orbital periods are now known for about 40 systems (Belczyński et al. 2000; Table 1). About half of these are eclipsing binaries whereas the reflection effect is responsible for the periodic light modulation in other systems. So far, only six systems—T CrB, CI Cyg, EG And, BD–21 3873, BF Cyg, and YY Her (Belczyński et al. 2000; Mikołajewska et al. 2002; Mikołajewska et al., this volume, p. 151)—show the ellipsoidal light variations of a tidally distorted red

giant. All systems with measured orbital period except R Aqr are S-type, and most of them have periods between ~ 200 – 1000 d (Table 1; Figure 3). Moreover, all systems with $P_{\text{orb}} \gtrsim 1000$ d, except for the yellow symbiotic CD–43 14302 contain very cool giants with spectral type $\gtrsim$ M6. The longest orbital period thus far estimated in symbiotic system is the 44 yr period of the symbiotic Mira R Aqr. The orbital periods of symbiotic Miras with thick dust (D-type) envelopes could be even larger. In general, the binary separations in D-type systems must be larger than the dust formation radius. Assuming a typical dust formation radius of $\gtrsim 5 \times R_{\text{Mira}}$, and $R_{\text{Mira}} \sim 1$ – 3 au, the minimum binary separation is $a \gtrsim 20$ au, and the corresponding binary period is $P_{\text{orb}} \gtrsim 50$ yr, for *any* D-type system. Thus the orbital period distribution seems to be the result of a compromise between the minimum binary separation and yet provides enough space for the evolved giant and the minimum mass accretion rate required for triggering the symbiotic phenomenon.

In a binary with $P_{\text{orb}} \gtrsim 1$ yr, the expected amplitude of radial velocity changes is rather low, $K_g \sim 5$ – 10 km s $^{-1}$. Nevertheless, modern observations with photon-counting detectors and new methods of analysis have allowed us to derive spectroscopic orbits from the radial velocities of the red giant for most of the bright symbiotic stars (Table 1). It is interesting that the vast majority of symbiotic stars have circular (or nearly circular, $e \lesssim 0.1$) orbits, and a significant eccentricity has been found only for BX Mon, KX TrA, CH Cyg, and CD–43 14304, which also have the longest orbital periods ($P_{\text{orb}} > 1000$ d). The eccentricity–period distribution for symbiotic stars is significantly different from the e – P_{orb} distribution in other binaries with late-type giants, which show a significant fraction of eccentric orbits among systems with $P_{\text{orb}} \lesssim 1000$ d (e.g., figure 9 of Jorissen & Mayor 1992).

In several cases, the motion of the hot component was also measured from either the blue absorption features or the optical and UV emission lines and broad emission line wings, and the masses of both components could be derived (Table 2). For some systems, the component masses were derived by combining the mass function for the red giant with the giant mass calculated from $v \sin i$ measurements and evolutionary tracks (e.g., Mürset et al. 2000). The distributions of mass ratios and the masses of the symbiotic binary components are shown in Figure 3b–d. All the systems, except BX Mon and the symbiotic recurrent novae T CrB and RS Oph, have a mass ratio, $q = M_g/M_h$, of between 2 and 4. The very high mass ratio for BX Mon is based on only two measurements of the radial velocities of the blue absorption lines and has to be confirmed by further observations covering at least one orbital period. The cool giant masses are all between 0.6 and $3.2 M_{\odot}$, with a peak around $1.6 M_{\odot}$ (the average is $1.7 \pm 0.1 M_{\odot}$). The masses of the hot component are all between 0.4 and $0.8 M_{\odot}$ (with the average at $0.53 \pm 0.05 M_{\odot}$) except those of AR Pav, T CrB, and RS Oph, and they fall into the range of masses expected for white dwarfs. In both symbiotic recurrent novae, the cool giant is the less massive component, and its mass, $M_g \lesssim 0.8 M_{\odot}$, is lower than that of any other symbiotic giant, whereas their hot components, with $M_h \sim 1.1$ – $1.3 M_{\odot}$, are the most massive, and their masses are sufficient for them to become Type Ia supernovae.

Finally, the direct measurements of hot component masses can be used to test the possible relation between white dwarf mass and orbital period (Miko-

Table 1. Orbital elements for symbiotic binaries

Star	P [d]	K_g [km s ⁻¹]	q (M_g/M_h)	γ_0 [km s ⁻¹]	e	T_0 [JD ¹]	$a_g \sin i$ [$R_\odot$]	$f(M)$ [$M_\odot$]
EG And ^[1]	482.6	7.3		−95.0	0	50804 ³	70	0.020
AX Per ^[2]	682.1	7.8	2.3	−117.4	0	50964	105	0.033
BD Cam ^[1]	596.2	8.5		−22.3	0.09	42794 ²	99.7	0.037
V1261 Ori ^[1]	642	7.5		79.7	0.07	46778 ³	95	0.028
BX Mon ^[1]	1401	4.3	6.7	29.1	0.49	49530	104	0.0076
^[1]	1259	4.6		29.1	0.44	49680	103	0.0092
SY Mus ^[1]	624.5	7.4		12.9	0	49082 ³	91	0.026
TX CVn ^[1]	199	5.7		2.3	0.16	45195 ²	22	0.004
RW Hya ^[1]	370.2	8.8		12.4	0	45072	65	0.026
^[1]	370.4	8.8		12.9	0	49512		0.026
BD−21 3873 ^[1]	281.6	10.6		203.9	0	49087 ³	59	0.035
T CrB ^[1]	227.57	23.9	0.6	−27.8	0	47919 ³	107	0.322
AG Dra ^[2]	549	5.9		−147.2	0	50775	64	0.0115
KX TrA ^[4]	1350	6.8	23	−123.7	0.29	51703	175	0.039
AE Ara ^[5]	812	5.4	4	−15.7	0	50217	87	0.0133
RS Oph ^[1]	455.7	16.7	0.35	−40.2	0	50154 ³	150	0.221
V343 Ser ^[3]	451.3	2.6		−5.63	0	50398 ³	23	0.0008
^[3]	450.5	2.7		−5.65	0.14	50575 ²	23.5	0.0009
FG Ser ^[2]	633.5	6.9		73.3	0	51031	87	0.022
AR Pav ^[6]	604.5	10.9	2.5	−68.3	0	48139	130	0.079
V443 Her ^[2]	594	2.5		−55.5	0	50197	30	0.0010
FN Sgr ^[7]	568.3	10.5	2.1	−53.7	0	50269	118	0.0689
BF Cyg ^[3]	757.2	6.7	3.6	−3.75	0	51395	100	0.0239
CH Cyg ^[1]	5700	4.9		−57.7	0.47	45086	478	0.045
^[1]	756.0	2.6		−60.6	0	46644	39	0.0014
^[1]	5292	4.8			0.06	45592 ²	500	0.060
CI Cyg ^[1]	855.3	6.7	3	18.4	0	45242	114	0.027
^[1]	853.8	6.7		15.0	0.11	50426	112	0.026
V1329 Cyg ^[3]	956.5	7.9	2.9	−23.1	0	51565	149	0.0481
CD−43 14304 ^[1]	1448	4.4		27.6	0	45929 ³	126	0.013
^[1]	1442	4.6		27.5	0.22	45560 ²	128	0.014
AG Peg ^[1]	816.5	5.3	4	−15.9	0	31668	84	0.012
^[1]	818.2	5.4		−15.9	0.11	46812	87	0.0135
Z And ^[2]	758.8	6.7		−1.8	0	50260	102	0.024
CD−27 8661 ^[1]	763.3	10.5		−5.5	0	49280 ³	158	0.092

T_0 : time of inferior spectroscopic conjunction of the giant; ¹Julian date = 2 400 000 + JD listed in table; ²time of the passage through periastron; ³time of maximum velocity.
References: ^[1]Belczyński et al. 2000, and references therein; ^[2]Fekel et al. 2000, and references therein; ^[3]Fekel et al. 2001, and references therein; ^[4]Ferrer et al. this volume, p. 117; ^[5]Mikołajewska et al., this volume, p. 147; ^[6]Quiroga et al. 2002, Quiroga et al., this volume, p. 490; ^[7]Brandi et al., this volume, p. 105.

Table 2. Mass estimates for symbiotic binaries

Star	P [d]	Ecl.	i [deg]	M_g [$M_\odot$]	M_h [$M_\odot$]	Comment
EG And	481	Y	90	1.5 ± 0.6	0.4 ± 0.1	ET
AX Per	680.8	Y	90	0.9 ± 0.2	0.37 ± 0.06	BA
			$\gtrsim 70$	$\lesssim 1.1$	$\lesssim 0.44$	
BX Mon	1401	Y	90	3.0 ± 1.5	0.45 ± 0.21	BA
			$\gtrsim 62$	$\lesssim 3.7$	$\lesssim 0.6$	
SY Mus	625	Y	90	1.3 ± 0.25	0.43 ± 0.05	ET
RW Hya	370.2	Y	90	1.6 ± 0.3	0.48 ± 0.06	ET
T CrB	227.57	N	~ 60	0.7 ± 0.2	1.2 ± 0.2	BM
KX TrA	1350	?	90	1.0 ± 0.3	0.41 ± 0.04	He II W
			135	2.7	1.2	SP
AE Ara	812	N	60	2.0 ± 1.2	0.51 ± 0.2	He II W
RS Oph	455.7	N	≤ 45	≥ 0.40	≥ 1.1	BA
FG Ser	650	Y	90	1.7 ± 0.7	0.60 ± 0.15	ET
AR Pav	604.5	Y	90	2.5 ± 0.6	1.0 ± 0.2	BA
			$\gtrsim 70$	$\lesssim 3$	$\lesssim 1.2$	
		Y	90	2.0 ± 0.5	0.87 ± 0.15	ET
FN Sgr	568.3	Y	90	1.4 ± 0.2	0.66 ± 0.08	BA
			$\gtrsim 70$	$\lesssim 1.7$	$\lesssim 0.8$	
BF Cyg	757.2	Y	90	1.8 ± 0.6	0.51 ± 0.1	UVEL
			$\gtrsim 70$	$\lesssim 2.2$	$\lesssim 0.6$	
CI Cyg	855.3	Y	90	1.3 ± 0.3	0.43 ± 0.04	He II EL
			≥ 79	≤ 1.6	≤ 0.52	
V1329 Cyg	956.5	Y	86	2.1 ± 0.5	0.74 ± 0.08	H I W, SP
AG Peg	816.5	N	$\lesssim 60$	$\gtrsim 1.8$	$\gtrsim 0.46$	He II EL

BA: blue absorption system; UVEL: ultraviolet emission lines; He II W: He II emission wings; H I W: H I emission wings; He II EL: He II emission line; SP: i from spectropolarimetry; ET: the cool giant mass from $v \sin i$ and evolutionary tracks (Mürset et al. 2000, and references therein; Schild et al. 2001); BM: light-curve synthesis combined with radial velocity curve (Belczyński & Mikołajewska 1998).

lajewska & Kenyon 1992a). The data from Table 2 are plotted in Figure 5 (left panel) and do not show any significant correlation with the orbital period. We must, however, note that the estimates for BX Mon and KX Tra, the only systems with longer periods, $P_{\text{orb}} > 1000$ d, are among the most uncertain results in Table 2.

4. The Cool Giant

Based on NIR spectra, Mürset & Schmid (1999) classified the cool giants in about 100 systems, and found that for S-type systems the spectral types cluster strongly between M3 and M6, with a peak at M5, whereas the distribution of symbiotic Miras peaks at spectral types M6 and M7. They also noticed a strong bias towards later spectral types when compared to red giants in the solar neighborhood, and that the frequency of Mira variables is higher among symbiotic giants. This predominance of very late, and thus more evolved, giants in symbiotic systems indicates that large radius and high mass loss from the

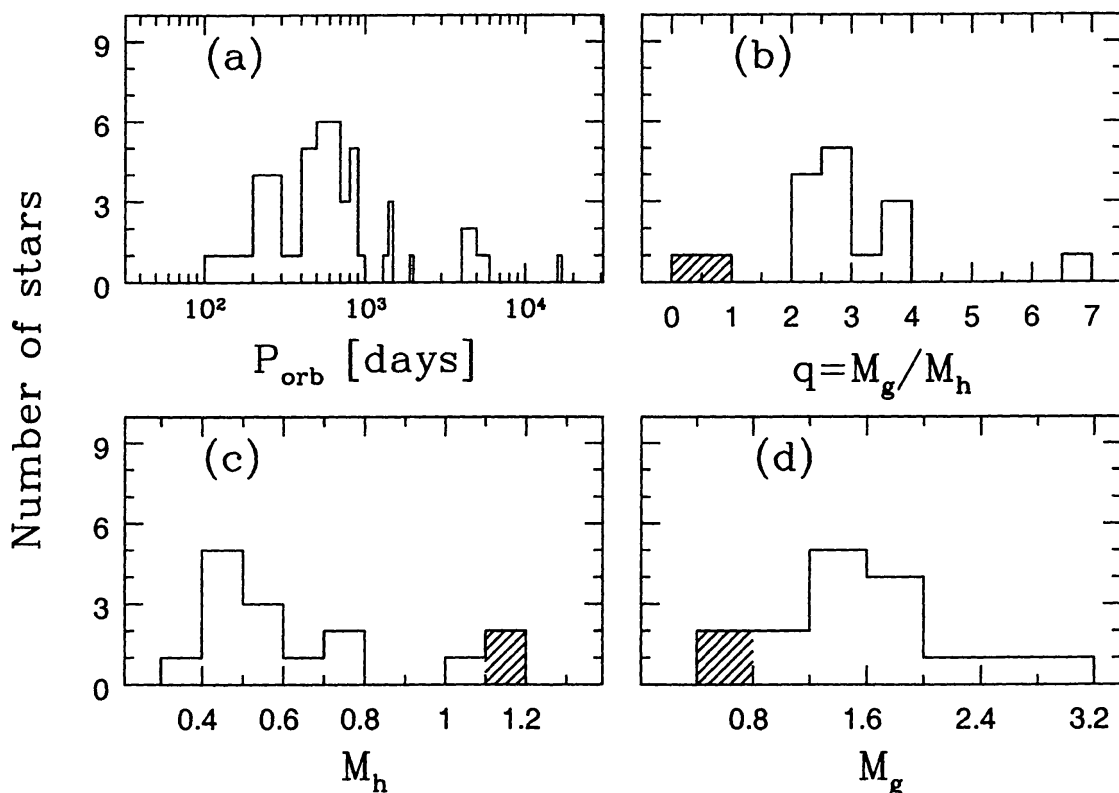


Figure 3. Distribution of a) binary periods, b) mass ratios, c) hot component masses, and d) cool giant masses. The shaded regions denote symbiotic recurrent novae.

cool giant are the key parameters for triggering the symbiotic phenomenon in binaries.

The process of mass transfer—Roche lobe overflow or stellar wind—is one of the fundamental questions in relation to symbiotic binaries. Although wind accretion obviously applies in the case of companions of symbiotic Miras, at least in some of the S-type systems, the cool giant could, in principle, fill its tidal lobe. Among the main evidence for the predominance of wind accretion is the fact that ellipsoidal light variations, characteristic of tidally distorted stars, are rarely observed for symbiotic stars. However, the general absence of such changes may in fact be due to the general lack of systematic searches for ellipsoidal variations in the red and NIR wavelength range, where the cool giant dominates the continuum. This is best illustrated by CI Cyg, BF Cyg, and YY Her, in which only the quiescent *VRI* and NIR light-curves show a modulation with half-orbital period as expected for ellipsoidal variability of the red giant, whereas the *UB* light-curves do not show clear evidence for such changes (Mikołajewska 2001; Mikołajewska et al., this volume, p. 151; Mikołajewska et al. 2002a). Interestingly, such changes have thus far been found only in systems with multiple outburst activity, whereas they are apparently absent in two non-eruptive systems, V433 Her and RW Hya (Mikołajewska et al. 2002b).

On the basis of on a sample of eight symbiotic systems with accurately determined orbits and estimated red giant radii, Schild et al. (2001) argued

that none of the symbiotic giants with known radii (derived from the measured $v \sin i$ in six cases and from eclipse contacts in AX Per) fills its Roche lobe. We note, however, that these authors did not include in their sample any systems with evident ellipsoidal changes. Similarly, Fekel, Hinkle, & Joyce (this volume, p. 113) derived the $v \sin i$ values and estimated the cool giant sizes for thirteen symbiotic systems. Their sample includes CI Cyg, BF Cyg, and T CrB; however, only CI Cyg has large Roche lobe filling factors $R_g/R_L \sim 0.8$, whereas the giants in both BF Cyg and T CrB, with $R_g/R_L \sim 0.3$ and ~ 0.4 , are far from filling their tidal lobes. A rotational velocity consistent with a synchronously rotating Roche lobe-filling giant was found in CI Cyg and AX Per (Kenyon et al. 1991; Mikołajewska & Kenyon 1992b). Recently, Orosz, & Hauschildt (2000) have shown that rotational broadening kernels for tidally distorted giants can be significantly different from analytic kernels, owing to a combination of the non-spherical shape of the giant and radical departure from a simple limb-darkening law. As a result, geometrical information inferred from $v \sin i$ measurements of cool giants in binary systems, and in particular in symbiotic stars, is probably biased. Summarizing, in our opinion the question of whether or not symbiotic giants, at least in some of the active S-type systems, are tidally distorted remains open as long as we do not have good NIR light-curves that would confirm or preclude such a possibility.

Mass loss rates for the symbiotic giants derived from either the cm and mm/sub-mm radio observations (Seaquist, Krogulec, & Taylor 1993; Mikołajewska, Ivison, & Omont 2002a; this volume, p. 478) or from the analysis of *IRAS* data (Kenyon, Fernandez-Castro, & Stencel 1988) are of order of $10^{-7} M_\odot \text{ yr}^{-1}$ and are systematically higher than those reported for single M giants, which again suggests that a high mass loss rate for the giant is essential for producing the symbiotic star.

Whitelock & Munari (1992) suggested that the symbiotic giants may be related to the metal-rich M stars found in the Galactic bulge and elsewhere, i.e., that they have low masses, $\sim 1 M_\odot$, and $Z \gtrsim Z_\odot$. They also noted that the mass loss rates of the symbiotic giants, although systematically greater than for the local bright giants, are similar to those of the bulge-like stars. Their findings, however, have not been confirmed by direct estimates of elemental abundances. In particular, Nussbaumer et al. (1988) found that the CNO abundance ratios deduced from UV emission lines for 24 symbiotic stars are best fitted by normal red giants. Moreover, carbon abundances and $^{12}\text{C}/^{13}\text{C}$ ratios have been obtained by fitting synthetic spectra to the observed first-overtone CO absorption features in *K* band spectra of seven northern and six southern symbiotic systems (Schmidt & Mikołajewska, this volume, p. 163, and references therein). In all cases, subsolar carbon abundances and $^{12}\text{C}/^{13}\text{C}$ ratios have been found which indicate that the symbiotic giants surveyed are indistinguishable from local normal M giants, in agreement with the abundance studies based on nebular emission lines.

5. The Hot Component and Activity

The typical hot components of symbiotic binaries appear to be quite hot ($T_{\text{eff}} \gtrsim 10^5 \text{ K}$) and luminous ($L_h \gtrsim 10^2\text{--}10^3 L_\odot$) and overlap in the same region in

the HR diagram as the central stars of planetary nebulae (Figure 4; Mürset et al. 1991; Mikołajewska, Acker, & Stenholm 1997). There is also no significant difference between the symbiotic hot components in our Galaxy and those in the Magellanic Clouds, except that the latter might be slightly hotter. It is hard to believe that such high luminosities are powered solely by accretion, as they would require accretion rates of $\sim$ a few times $10^{-6} M_{\odot} \text{ yr}^{-1}$. Although symbiotic giants have high mass loss rates ($\sim 10^{-7} M_{\odot} \text{ yr}^{-1}$ and more), in most cases they interact via stellar winds instead of Roche lobe overflow, and the expected accretion rate is an order of magnitude lower. The situation radically changes if the symbiotic white dwarfs burn hydrogen-rich material as they accrete it. An accretion rate of the order of $10^{-8} M_{\odot} \text{ yr}^{-1}$ is then sufficient to power the typical hot component with $M_h \sim 0.5 M_{\odot} \text{ yr}^{-1}$.

According to the theoretical studies, the hydrogen-shell burning luminosity should be a function of the underlying core mass. Unfortunately, there is no unique core mass–luminosity relationship for accreting white dwarfs because it depends significantly on the thermal history of the white dwarf: generally, hot white dwarfs must have larger masses than the cooler ones to reach the same luminosity during the hydrogen-burning phase (Iben & Tutukov 1996). However, for symbiotic systems with known component masses (Table 2) we are in a position to determine which of the theoretical mass–luminosity relations is applicable in their case. The positions of the symbiotic hot components in the luminosity versus mass plane together with different mass–luminosity relationships are plotted in Figure 5 (right panel). The symbiotic white dwarfs seem to cluster around the mass–luminosity relations for stars leaving the AGB with a CO core and those leaving the RGB with a degenerate He core for the first time. It is possible that the white dwarf descendant of the more massive component could still be hot at the onset of mass transfer from the less massive red giant.

The hot components in many symbiotic systems show intrinsic variability (Figure 2). On the basis of this activity, we distinguish between ordinary or classical symbiotic stars, which show occasional 1–3 mag eruptions on timescales from months to a few years (such as Z And, CI Cyg, and AG Dra), and symbiotic novae that have undergone a single outburst of several magnitudes lasting for tens of years. Figure 6 shows the outburst evolution of selected symbiotic systems. The outburst behavior of AG Peg and RX Pup is consistent with a thermonuclear nova eruption. In both systems the outburst develops very slowly: the rise to maximum takes months, and the decline to the pre-outburst stage lasts tens of years. AG Peg is the record-holder among symbiotic novae: its eruption began in the 1850s, and the hot component maintained a constant luminosity, $L_h \sim 3000 L_{\odot}$, for at least 100 years. The evolution of RX Pup was much faster, the constant-luminosity (plateau) phase lasted for only 11 years, and the maximum plateau luminosity, $\langle L_h \rangle \sim 15000 L_{\odot}$, was about five times higher than that of AG Peg. These differences in outburst evolution can be accounted for by different white dwarf masses (higher in RX Pup than that in AG Peg). RX Pup is also a possible recurrent nova.

Unfortunately, the thermonuclear models do not account for the multiple outburst activity of Z And, AG Dra (Figure 6), and other classical systems. In most of these, the hot component maintains a roughly constant luminosity, whereas its effective temperature varies from $\sim 10^5$ to $\sim 10^4$ K, and Z And is a

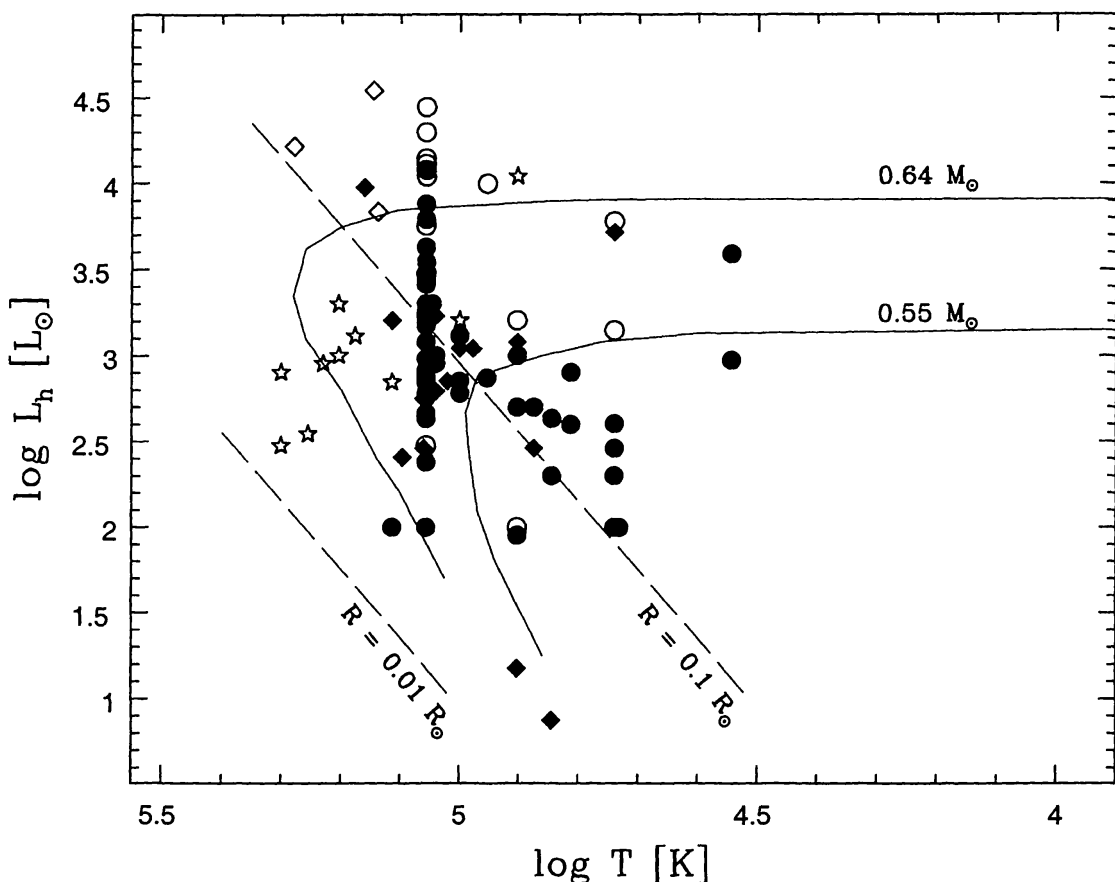


Figure 4. The hot components of symbiotic stars in the HR diagram. Data from Mürset et al. (1991) and Mikołajewska, Acker, & Stenholm (1997) are plotted as diamonds and circles, whereas filled and open symbols represent the Galactic S- and D-types, respectively. Stars correspond to the symbiotic stars in the Magellanic Clouds (Morgan 1992, 1996). The solid curves are the evolutionary tracks from Schönberner (1989), and the dashed lines correspond to constant radii.

good example of such an outburst. In contrast to this behavior, the luminosity of the hot component of AG Dra increases by a factor of ~ 10 , whereas its temperature increases during the early stages of each eruption and then decreases as the outburst continues. A possible and promising explanation of this activity involves changes in mass transfer and/or accretion disk instabilities. There are many arguments in favor of such an explanation. First, the hot companions of all symbiotic giants that fill or nearly fill their tidal lobes show multiple outburst activity. Moreover, even if the giant does not fill its Roche lobe, its wind is probably focused towards the secondary and/or towards the orbital plane, which would facilitate accretion disk formation (e.g., Gawryszczak, Mikołajewska, & Różyczka 2002; Mastrodemos & Morris 1999).

The timescales for these eruptions are very similar to those of the hot component luminosity changes (high and low states) in symbiotic recurrent novae (T CrB, RS Oph, and RX Pup) between their nova eruptions and the accretion-powered system of CH Cyg and MWC 560. During bright phases, in all these systems the optical data indicate the presence of a relatively cool B/A/F-type

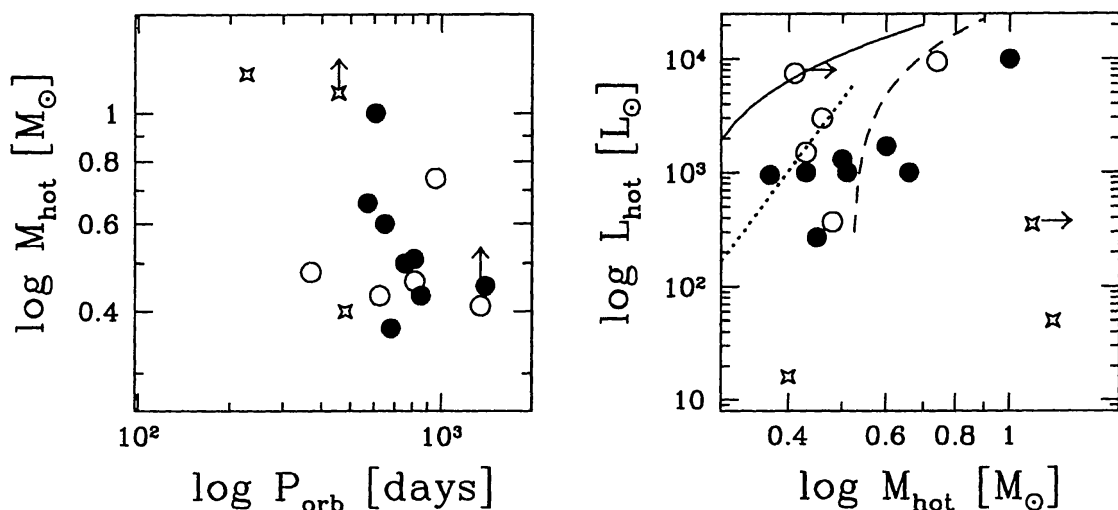


Figure 5. Mass–orbital period relation (*left*), and mass–luminosity relation (*right*) for the hot components. Closed symbols represent systems with Z And-type multiple outburst activity; open circles: non-eruptive systems and symbiotic novae; stars: accretion-powered systems, namely symbiotic recurrent novae at quiescence and EG And. The upper solid curve is the mass–plateau luminosity relationship for accreting cold white dwarfs (eq. 6 of Iben & Tutukov 1996), the dashed curve is the Paczyński–Uus relation (e.g., Paczyński 1970) for AGB stars with a CO core, and the dotted one is the relation for helium dwarfs (eq. 7 of Iben & Tutukov 1996).

shell source while the UV/optical emission lines require a much hotter source with a roughly comparable luminosity (Mikołajewska et al. 2002c). A similar double-temperature structure of the hot component is observed in Z And-type systems during their outbursts. In AX Per, AR Pav, FN Sgr, and possibly other active systems, the A/F-type absorption lines trace the orbit of the hot component and are probably formed in a geometrically and optically thick accretion disk and in a gas stream (Mikołajewska & Kenyon 1992b; Quiroga et al. 2002; Quiroga et al., this volume, p. 490; Brandi et al., this volume, p. 105). Other evidence for the presence of accretion disks in active symbiotic stars includes the ellipsoidal shape of the blue continuum source during outburst, presence of secondary periods (always 10–20% shorter than P_{orb}) in the outburst light-curves, similar to the superhumps in CVs (Mikołajewska et al. 2002b); bipolar geometry of outflows associated with the hot component activity in CI Cyg, AG Dra, and other systems (Mikołajewska & Ivison 2001; Mikołajewska 2002; Tomov, Munari, & Maresse 2000). It is possible that both the activity of classical Z And-type symbiotic stars and the high and low states of accretion-powered systems are related to the presence of unstable accretion disks, with the only difference that the hot component in the former burns the accreted hydrogen more or less stably, but not in the latter.

Finally, Mikołajewska & Kenyon (1992a) argued that the outburst evolution of CI Cyg and AX Per are best explained by the presence of an unstable thick disk around a low mass main sequence accretor. We must, however, note that the quiescent characteristics of their hot components are consistent with a hot and luminous stellar source powered by thermonuclear burning. In partic-

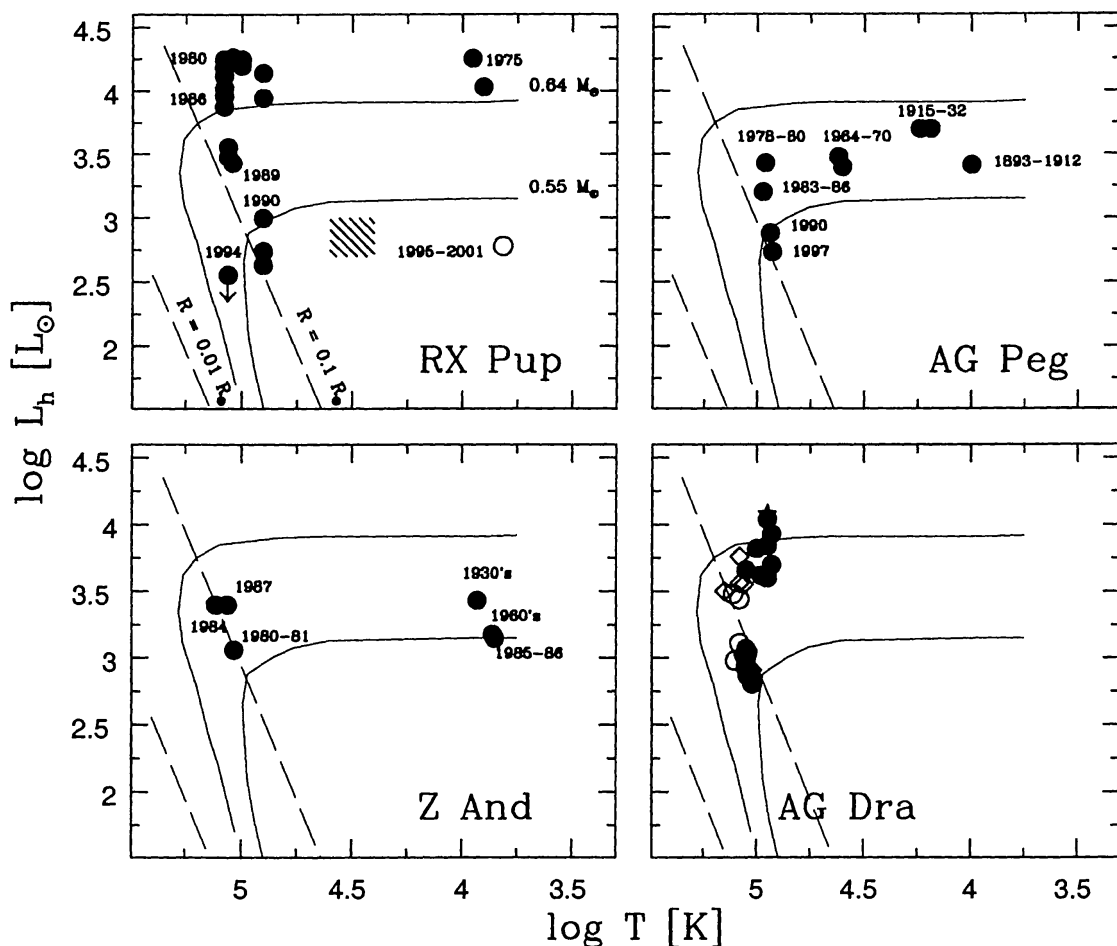


Figure 6. Outburst evolution of the hot component of the symbiotic novae RX Pup and AG Peg, Z And, and the yellow symbiotic AG Dra (Mikołajewska et al. 1999; Kenyon et al. 1993; Kenyon, Proga, & Keyes 2001; Mikołajewska & Kenyon 1996; Mikołajewska et al. 1995.)

ular, they fall in the same region in the HR diagram as the hot components of other symbiotic stars (Figure 4). Their outburst evolution is also very similar to that of Z And, FN Sgr, and other multiple-outburst systems, which apparently points to a common mechanism for the outburst and the nature of their hot component. Unfortunately, it will remain uncertain as long as we do not have a good theoretical model accounting for both their quiescent and outburst characteristics.

6. Concluding Remarks

The nature of the cool giant plays the leading role in the symbiotic phenomenon because it constrains the size of the binary system, which must have enough room for a red giant and yet transfer enough material to the companion and give rise to the symbiotic apparition. As a result, we have two distinct classes: the S type with normal giants and orbital periods of $\lesssim 15$ yr, and the D type with Mira primaries and periods of $\gtrsim 50$ yr.

Most S-type systems have $P_{\text{orb}} \sim 200\text{--}900$ d and circular orbits. A typical S-type system consists of a low mass M3–6 giant ($\langle M_g \rangle \sim 1.7 M_\odot$) transferring material to a hot, $\sim 10^5$ K, white dwarf companion ($\langle M_h \rangle \sim 0.5 M_\odot$). Although most symbiotic stars seem to interact by wind-driven mass loss, at least some of the systems with multiple outburst activity may contain Roche lobe-filling (or nearly filling) giants.

Symbiotic M giants have subsolar carbon and $^{12}\text{C}/^{13}\text{C}$ abundances and are indistinguishable in this respect from local M giants. They have, however, systematically higher mass loss rates than single giants, which suggests that a high mass loss rate for the giant is essential for triggering the symbiotic activity. The chemical abundances derived from nebular emission lines suggest that the main body of symbiotic nebulae is formed from material lost in the giant wind, whereas the hot companion is responsible for its ionization and excitation.

The typical hot component of a symbiotic binary appears to be a luminous, $\sim 100\text{--}10^4 L_\odot$, and hot, $\sim 10^5$ K, white dwarf powered by thermonuclear burning of the accreted hydrogen. The position of the symbiotic white dwarfs in the mass–luminosity plane indicates that they could be hot and luminous at the onset of mass transfer and symbiotic activity. Although we still do not have a satisfactory explanation of multiple outburst activity, there is increasing evidence that it may be related to the presence of unstable accretion disks.

Acknowledgments. This research was partly funded by KBN Research Grant No. 5 P03D 019 20.

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Discussion

Whitelock: That was a very nice introduction to the meeting. I would like to comment on the point you made early in your talk that the spectral types of the symbiotic stars were later, and the mass-loss greater, than in field stars. I think there is a problem with the field stars that the comparison is being made with. They are bright local stars and are higher in mass than the typical M giants in symbiotic stars, so the difference is not surprising. We showed some while ago (Whitelock & Munari 1992) that if you compare with low mass M giants in the disk or the bulge there are no significant differences between symbiotic and field M giants.

Mikołajewska: As far as I remember, the main conclusion of your paper was that the symbiotic giants seem to be related to metal-rich M giants, which also have higher mass loss. However, neither the dynamical mass estimates, which suggest an average mass of $\sim 1.7 M_{\odot}$ (i.e., somewhat higher than $\sim 1 M_{\odot}$ suggested for these high Z giants), nor direct estimates for elemental abundances (see the poster presented by Schmidt & Mikołajewska [this volume, p. 163]) support your conclusion. I think that the problem needs further study, in particular, chemical abundance analysis and accurate mass loss rate estimates for a large sample of symbiotic giants.

Whitelock: You mentioned an orbital period of ~ 5000 days for RT Ser. Is that correct and what is the mass transfer mechanism in such a long P system (noting that the M star is *not* a Mira variable)?

Mikołajewska: Yes, it is correct. There will be a contributed talk later today which will discuss the photometric variability of RT Ser with a period of about 12 years (~ 4500 days).

Wallerstein: Do not forget R Aquarii with a period of 43 or 44 years.

Mikołajewska: Yes, I remember that one. My table, however, contains only systems for which we have a spectroscopic orbit for the red giant.

Jorissen: In the (L_h, M_h) diagram, AR Pav appears to be somewhat outlying. On the other hand, there have been claims in the literature that AR Pav has a main sequence companion.

Mikołajewska: As far as I remember the main argument for the MS companion in AR Pav was the mass, $M_h \sim 2-4 M_{\odot}$, derived from light-curve analyses and radial velocities of emission lines. I must, however, emphasize that the recent dynamical mass estimates for both components are compatible with a white dwarf. Despite the relatively high mass and luminosity, the hot component of AR Pav and its outburst behavior are similar to those of CI Cyg, Z And, FN Sgr, and other multiple-outburst systems, which, in my opinion, points to the common nature of their hot component. However, the question of whether it is a disk-accreting main sequence star or a quasi-steady hydrogen-burning white dwarf is open so long as we have no good theory to distinguish between these two possibilities.